SMART INDIA HACKATHON 2026

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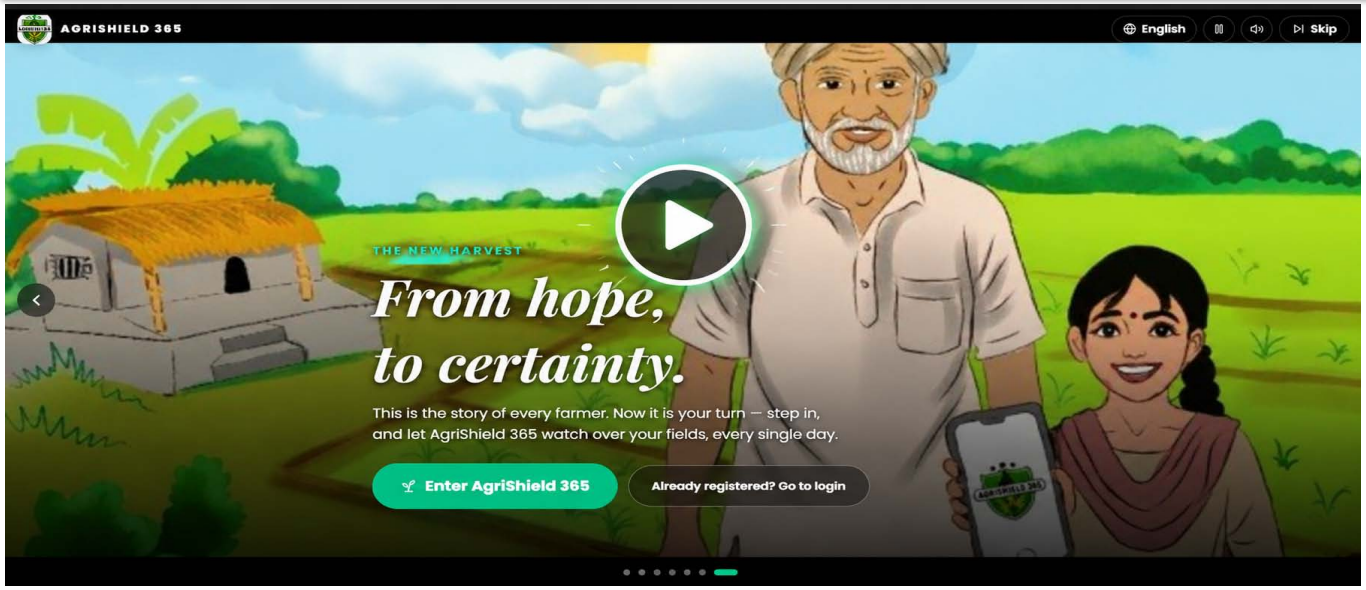
- Problem Statement ID – SIH26131
- Problem Statement Title - Early detection and management of crop diseases and pest infestations
- Theme – Agriculture ,Food Tech & Rural Development
- PS Category - Software
- Team ID - 166880
- Team Name (Registered on portal) - RAVEN ROOTS

AGRISHIELD 365
Every Farmer, Every Season , Protected

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Solution Mechanics: How It Solves the Problem

Brief Walkthrough of Problem and how AgriShield-365 solve it



Uniqueness of AGRISHIELD-365

1) Preventive, not Reactive (Predicts future in place of waiting for it)
2) Two-Brain Architecture(Local + cloud brain)
3) Offline-first farmer safety (PWA first priority)
4) Multi-lingual and farmer first design
5) Smart irrigation with crop science (Jal-Rakshak Module)
6) Market awareness included
7) Explainable advisories (Doesn't just say 'high risk' , it explains why)

SERVICES for our Farmer's benefits



Live App Link :-
https://agrishield365.web.app/
YouTube Video Link :-
https://youtu.be/tJPDBoMmfGI
(Full Detailed Walkthrough)

Scan

AgriShield 365
Scan to open the live app
agrishield365.web.app

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TECHNICAL APPROACH

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Methodology and Implementation Process with Tech Stack

Two Brain, Never Blocking Design

Edge Intelligence

PWA on the Farmer's Phone featuring a local rule engine, re-silient data, and AgriSaathi assistant.

Cloud Enrichment

FastAPI / Render backend with synchronized logic, advanced vision, and persistent storage.

Global Infrastructure

Data sources, secu-rity, and hosting via Open-Meteo, Google Gemini, and Firebase.

Data flows through emerald and navy dashed paths, connecting real-time weather from Open- Meteo and domain logic from ICAR/TNAU/FAO-56 to provide a seamless agricultural monitoring experience.

Dark Themed Dashboard



React 19

Modern web interface framework

Firebase

Real-time cloud database services

AGRISHIELD 365

FastAPI

High-performance backend API framework

Google Gemini Vision

Advanced im-age analysis engine

Open-Meteo

Real-time weather data integration

ICAR/TNAU/FAO-56

Domain specific agricultural logic

Revolutionizing Agriculture Through Data

- Farmer-Centric Design
- Real-Time Data
- Holistic Support
- Secure Authentication

Light Themed Dash-board



Made with Napkin

FEASIBILITY AND VIABILITY

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Feasibility & viability of the idea, with strategies to overcome challenges

Turning Data into Action • Supporting Farmers • Building a Sustainable Future

AgriShield 365
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agrishield365.web.app

FEASIBILITY

- **Reuses Free Infrastructure**
  Firebase, Open-Meteo & PWA hosting
- **Scalable**
  Same app serves one farm or a whole district
- **Cost-Effective**
  Minimal running cost per farmer
- **Authority-Ready**
  Built for agriculture departments & insurers

VIABILITY

- **Proven Concept**
  Working live app with real farmer flows
- **Builds Public Trust**
  Advice grounded in ICAR/TNAU/FAO
- **Accountability**
  Verifiable alerts and claim records
- **Engagement**
  Story onboarding, voice & local languages

CHALLENGES → STRATEGIES

Patchy village internet

→ Offline-first PWA; cloud only enriches

Cloud AI quota / cost

→ Local engine + offline symptom checker never depend on Gemini

Low digital literacy

→ 4 languages, voice read-aloud + voice questions

Advice trust

→ ICAR/TNAU/FAO grounding + KVK disclaimer

SMS needs billing

→ Premium voice/SMS alerts funded via ₹99/month plan

AGRISHIELD 365 BUSINESS MODEL

4 Revenue Streams

Free core alerts for every farmer, funded by premium, B2G, B2B & fintech partners

B2C Freemium

Free core alerts; AI leaf scan & premium voice/SMS alerts at ₹99/month

B2G & Insurance

Verifies PMFBY claims & water audits — rapid validation of genuine crop-failure claims

B2B Agribusiness

Anonymised disease-outbreak heatmaps to pre-position organic bio-fungicides

Fintech Partners

Micro-credit scoring; low-interest crop loans for farmers with crop-loss history

RAVEN ROOTS

AgriShield 365

Scan to use

IMPACTS AND BENEFITS

AgriShield365 — the Boon of 21st-century India

Estimates grounded in CICR, FAO-56, IMD/TNAU

AGRISHIELD 365

SIH

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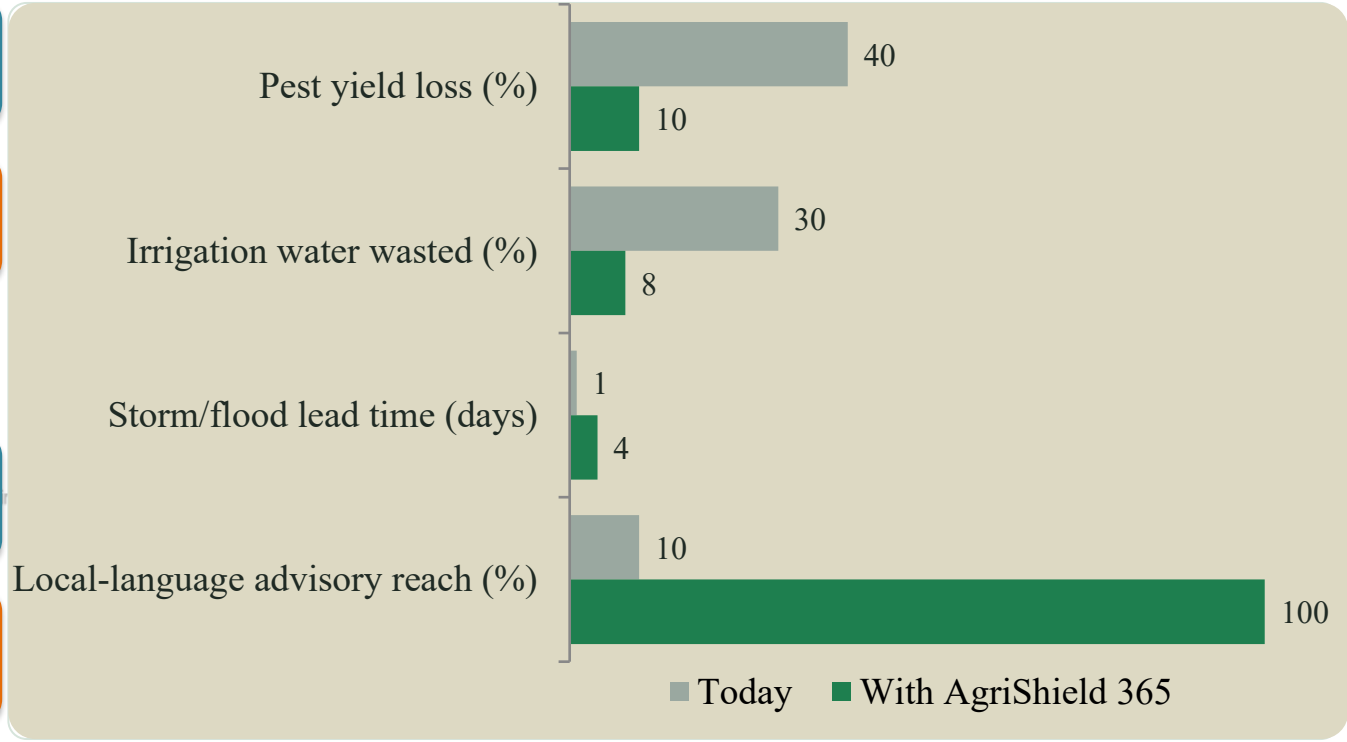
Who Benefits?

- Farmers
- Agri-Departments
- Water Managers
- Community

Why it Matters?

- Protect Crops
- Reduces Irrigation
- Alerts Timely
- Local Languages

POTENTIAL IMPACT ON INDIA'S AGRICULTURE



BREAKTHROUGH

Major Future Plan

KhetiScan — Geospatial Canopy Surveillance

Satellite NDVI/NDWI (Sentinel-2 / Landsat) spots pest-attacked & waterlogged zones across whole districts before field visits.

- *Smart Technology.*
- *Safer Crops.*
- *Sustainable Future.*

RESEARCH AND REFERENCES

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AGRISHIELD-365 is built upon these primary research sites and references

AgriShield 365

Pillars of AGRISHIELD 365



ICAR- Research for all 3 crops in details



TNAU- Research for 8 diseases of all 3 crops



FAO- Research for Kc values



Gov. of Maharashtra (Agriculture Dept.) mainly used for cotton-related research



Open Meteo API- Research for Weather Data



JalRakshak- Smart Irrigation (acts as "The Water Protector")

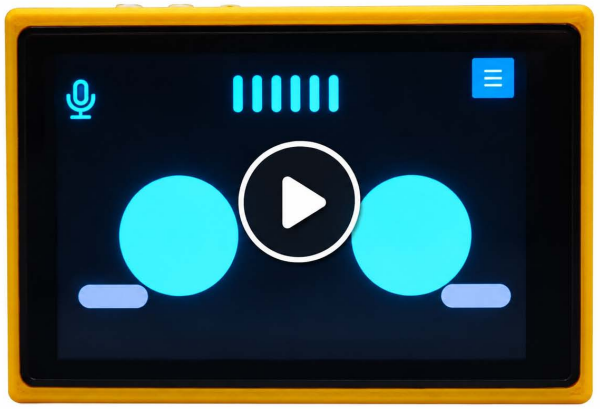


AgriSaathi- AI Assistant (turns app into a conversation)

Report Link: https://docs.google.com/document/d/12EoHv-kmXrN7Ubei2lQQ972usTdcUed9/edit?usp=sharing&ouid=105708397499206098346&rtpof=true&sd=true

Introducing LILLY : Our AGRISHIELD-365 deployed as Smart Farming Assistant

YouTube video link: https://youtu.be/a5vdY3kb6ZQ



Why hardware as a "Future plan" in a Software project ?

a) 24×7 –It works all day all night relentlessly.

b) his plot's data — sensors measure *your* field's leaf wetness, soil, canopy — not district-level downloaded weather

c) Acts, not just advises — can hold the pump itself and verify every liter of water saved.